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Heterogenous bonding layers for direct semiconductor bonding

Abstract

A first semiconductor device and a second semiconductor device may be directly bonded using heterogeneous bonding layers. A first bonding layer may be formed on the first semiconductor device and the second bonding layer may be formed on the second semiconductor device. The first bonding layer may include a higher concentration of hydroxy-containing silicon relative to the second bonding layer. The second bonding layer may include silicon with a higher concentration of nitrogen relative to the first bonding layer. An anneal may be performed to cause a dehydration reaction that results in decomposition of the hydroxy components of the first bonding layer, which forms silicon oxide bonds between the first bonding layer and the second bonding layer. The nitrogen in the second bonding layer increases the effectiveness of the dehydration reaction and the effectiveness and strength of the bond between the first bonding layer and the second bonding layer.

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Background/Summary

RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/446,549, filed Aug. 31, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Bonding in the semiconductor industry is a technique that may be used to form stacked semiconductor devices and three-dimensional integrated circuits. Some examples of bonding include wafer to wafer bonding, die to wafer bonding, and die to die bonding.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a diagram of an example environment in which systems and/or methods described herein may be implemented.

(3) FIGS. 2A-2G are diagrams of one or more example operations involved in manufacturing an example semiconductor device described herein.

(4) FIG. 3 is a diagram of example components of one or more tools and/or devices of FIG. 1 described herein.

(5) FIGS. 4 and 5 are flowcharts of example processes for joining semiconductor devices described herein.

(6) FIGS. 6A-6D are diagrams of one or more example operations involved in manufacturing an example semiconductor device described herein.

(7) FIG. 7 is a flowchart of an example process for joining a semiconductor device described herein.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(9) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(10) Various bonding techniques may be used to bond a first semiconductor device with a second semiconductor device, such as direct bonding, chemically activated bonding, plasma activated bonding, anodic bonding, eutectic bonding, glass frit bonding, adhesive bonding, thermo-compressive bonding, and reactive bonding, among other examples. Some bonding techniques involve the use of a plasma pretreatment technique. Plasma pretreatment includes using a plasma (e.g., a nitrogen-based plasma or another type of plasma) to pretreat one or more bonding films to

promote adhesion between the first semiconductor device and the second semiconductor device. Such bonding techniques may include several costly and time-consuming processing techniques.

(11) Some implementations described herein provide techniques and apparatuses for direct bonding of two semiconductor devices using heterogeneous bonding layers. In some implementations, a first bonding layer may be formed on a first semiconductor device and a second bonding layer may be formed on a second semiconductor device that is to be bonded or joined to the first semiconductor device. Each bonding layer may include a silicon-containing material. The first bonding layer may include a higher concentration of hydroxy-containing silicon (silicon bonded with a hydroxy, which may include an oxygen atom bonded to a hydrogen atom (OH)) and a lower concentration of nitrogen relative to the second bonding layer. The second bonding layer may include silicon with a higher concentration of nitrogen relative to the first bonding layer.

(12) A dry anneal may be performed to bond or fuse the first bonding layer and the second bonding layer. A dehydration reaction occurs between the first bonding layer and the second bonding layer during the dry anneal. The dehydration reaction results in decomposition of the hydroxy components of the first bonding layer, which forms silicon oxide bonds between the first bonding layer and the second bonding layer. Here, the silicon and the oxygen in silicon-hydroxy (Si—OH) of the first bonding layer bonds with the silicon in the second bonding layer to form Si—O—Si bonds, with water forming a byproduct of the dehydration reaction (e.g., the hydrogen and oxygen atoms in the Si—OH form H₂O). In this way, the first semiconductor device and the second semiconductor device may be directly bonded without the use of a plasma pretreatment process, which decreases a cost and a complexity of the bonding process. Moreover, the nitrogen in the second bonding layer increases the effectiveness of the dehydration reaction and, therefore, the effectiveness and strength of the bond between the first bonding layer and the second bonding layer.

(13) FIG. 1 is a diagram of an example environment **100** in which systems and/or methods described herein may be implemented. As shown in FIG. 1, environment **100** may include a plurality of semiconductor processing tools **102-112** and a wafer/die transport tool **114**. The plurality of semiconductor processing tools **102-112** may include a deposition tool **102**, an exposure tool **104**, a developer tool **106**, an etching tool **108**, a planarization tool **110**, an annealing tool **112**, and/or another type of semiconductor processing tool. The plurality of semiconductor processing tools **102-112** included in example environment **100** may be included in a semiconductor clean room, a semiconductor foundry, a semiconductor processing and/or manufacturing facility, and/or the like.

(14) The deposition tool **102** is a semiconductor processing tool that includes a semiconductor processing chamber and one or more devices capable of depositing various types of materials onto a substrate. In some implementations, the deposition tool **102** includes a spin coating tool that is capable of depositing a photoresist layer on a substrate such as a wafer. In some implementations, the deposition tool **102** includes a chemical vapor deposition (CVD) tool such as a plasma-enhanced CVD (PECVD) tool, a high-density plasma CVD (HDP-CVD) tool, a sub-atmospheric CVD (SACVD) tool, an atomic layer deposition (ALD) tool, a plasma-enhanced atomic layer deposition (PEALD) tool, or another type of CVD tool. In some implementations, the deposition tool **102** includes a physical vapor deposition (PVD) tool, such as a sputtering tool or another type of PVD tool. In some implementations, the example environment **100** includes a plurality of types of deposition tools **102**.

(15) The exposure tool **104** is a semiconductor processing tool that is capable of exposing a photoresist layer to a radiation source, such as an ultraviolet light (UV) source (e.g., a deep UV light source, an extreme UV light source, and/or the like), an x-ray source, and/or the like. The exposure tool **104** may expose a photoresist layer to the radiation source to transfer a pattern from a photomask to the photoresist layer. The pattern may include one or more semiconductor device layer patterns for forming one or more semiconductor devices, may include a pattern for forming

one or more structures of a semiconductor device, may include a pattern for etching various portions of a semiconductor device, and/or the like. In some implementations, the exposure tool **104** includes a scanner, a stepper, or a similar type of exposure tool.

(16) The developer tool **106** is a semiconductor processing tool that is capable of developing a photoresist layer that has been exposed to a radiation source to develop a pattern transferred to the photoresist layer from the exposure tool **104**. In some implementations, the developer tool **106** develops a pattern by removing unexposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by removing exposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by dissolving exposed or unexposed portions of a photoresist layer through the use of a chemical developer.

(17) The etching tool **108** is a semiconductor processing tool that is capable of etching various types of materials of a substrate, wafer, or semiconductor device. For example, the etching tool **108** may include a wet etching tool, a dry etching tool, and/or the like. In some implementations, the etching tool **108** includes a chamber that is filled with an etchant, and the substrate is placed in the chamber for a particular time period to remove particular amounts of one or more portions of the substrate. In some implementations, the etching tool **108** may etch one or more portions of the substrate using a plasma etch or a plasma-assisted etch, which may involve using an ionized gas to isotopically or directionally etch the one or more portions.

(18) The planarization tool **110** is a semiconductor processing tool that is capable of polishing or planarizing various layers of a wafer or semiconductor device. For example, a polishing device may include a chemical mechanical polishing (CMP) device and/or another type of polishing device. In some implementations, a polishing device may polish or planarize a layer of deposited or plated material.

(19) The annealing tool **112** is a semiconductor processing tool that includes a semiconductor processing chamber and one or more devices capable of heating a semiconductor device. For example, the annealing tool **112** may include a rapid thermal anneal (RTA) tool, a dry annealing tool, or another type of annealing tool that is capable of heating a semiconductor device to cause a reaction between two or more materials or gasses, to cause a material to decompose, to bond two or more semiconductor devices, and/or the like.

(20) Wafer/die transport tool **114** includes a mobile robot, a robot arm, a tram or rail car, and/or another type of device that are used to transport wafers and/or dies between semiconductor processing tools **102-112** and/or to and from other locations such as a wafer rack, a storage room, and/or the like. In some implementations, wafer/die transport tool **114** may be a programmed device to travel a particular path and/or may operate semi-autonomously or autonomously.

(21) The number and arrangement of devices shown in FIG. **1** are provided as one or more examples. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIG. **1**. Furthermore, two or more devices shown in FIG. **1** may be implemented within a single device, or a single device shown in FIG. **1** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of the environment **100** may perform one or more functions described as being performed by another set of devices of the environment **100**.

(22) FIGS. **2A-2G** are diagrams of one or more example operations **200** involved in manufacturing an example device described herein. The device may include a logic device, a memory device, a finFET, a MOSFET, an integrated circuit, a processor, a sensor, another type of semiconductor or electronic device, or a portion thereof. The example operations **200** illustrated and described in connection with FIGS. **2A-2G** may be performed as part of a bonding process to bond two or more semiconductor devices of the device.

(23) As shown in FIG. **2A**, the device may include a first semiconductor device **202**. The first semiconductor device **202** may include a first silicon layer **204**, an undoped silicate glass (USG) layer **206** provided on the first silicon layer **204**, first metal contacts **208** formed in the USG layer

206, and a passivation layer **209** formed on the USG layer **206** and the first metal contacts **208**. The first semiconductor device **202** may include a semiconductor wafer, a semiconductor die, and/or the like.

(24) The first silicon layer **204** may include a silicon wafer sliced from a silicon crystal ingot grown as a cylinder. The first silicon layer **204** may include an electrical conductivity value falling between that of a conductor, such as metallic copper, and an insulator, such as glass. The first silicon layer **204** may be replaced with other materials, such as germanium, gallium arsenide, silicon germanium, and/or the like.

(25) The USG layer **206** may include an undoped silicate glass that protects and isolates elements of the first semiconductor device **202**. The USG layer **206** may include a high deposition rate at low temperatures, and may include similar properties to silicon dioxide. The USG layer **206** may be utilized as an insulator and passivation layer in multilevel interlevel dielectric devices (e.g., to electrically insulate the first metal contacts **208** from other components of the first semiconductor device **202**). In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. **1**, may be utilized to form the USG layer **206** on a top surface of the first silicon layer **204**. For example, deposition tool **102** may perform a PECVD operation, an HDP-CVD operation, an SACVD operation, an ALD operation, a PVD operation, or another deposition operation to deposit the USG layer **206** on the top surface of the first silicon layer **204**.

(26) The first metal contacts **208** may include a conductive metal, such as titanium, cobalt, tungsten, aluminum, copper, ruthenium, iridium, and/or the like. In some implementations, the first metal contacts **208** may be formed within openings formed in the USG layer **206**. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. **1**, may be utilized to perform a deposition operation that forms the first metal contacts **208** in the openings of USG layer **206**. In some implementations, a plating tool is used to form the first metal contacts **208** in the USG layer **206**. In these examples, the plating tool may perform a plating operation to form the first metal contacts **208**. Plating may include applying a voltage across an anode formed of a plating material and a cathode (e.g., a substrate). The voltage causes a current to oxidize the anode, which causes the release of plating material ions from the anode. These plating material ions form a plating solution that travels through a plating bath toward the substrate (e.g., the USG layer **206**). The plating solution reaches the USG layer **206** and deposits plating material ions into the openings in the USG layer **206** to form the first metal contacts **208**.

(27) In some implementations, one or more semiconductor processing tools of the environment **100**, described above in connection with FIG. **1**, may be utilized to form the openings in the USG layer **206**, prior to formation of the first metal contacts **208** in the openings. For example, the deposition tool **102** may form a photoresist layer on the USG layer **206**, the exposure tool **104** may expose the photoresist layer to a radiation source to pattern the photoresist layer, the developer tool **106** may develop and remove portions of the photoresist layer to expose the pattern, and the etching tool **108** may etch the one or more portions of the USG layer **206** to form the openings in the USG layer **206**. In some implementations, a photoresist removal tool removes the remaining portions of the photoresist layer (e.g., using a chemical stripper and/or another technique) after the etching tool **108** etches the USG layer **206**.

(28) The passivation layer **209** may include an oxide material (e.g., a metal oxide) that is inert and does not change semiconductor properties as a result of interaction with air or other materials in contact with the passivation layer **209**. The passivation layer **209** may allow electricity to reliably penetrate to conducting layers provided below the passivation layer **209**, and to overcome surface states that prevent electricity from reaching the conducting layers. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. **1**, may be utilized to form the passivation layer **209** on the top surface of the USG layer **206** and the first metal contacts **208**.

(29) As further shown in FIG. **2A**, the device may include a second semiconductor device **210**. The

second semiconductor device **210** may include a second silicon layer **212**, a doped layer **214** provided on the second silicon layer **212**, an epitaxial layer **216** formed on the doped layer **214**, an intermetal dielectric (IMD) layer **218** formed on the epitaxial layer **216**, second metal contacts **220** formed in the IMD layer **218**, and a passivation layer **221** formed on the IMD layer **218** and the second metal contacts **220**. The second semiconductor device **210** may include a semiconductor wafer, a semiconductor die, and/or the like.

(30) The second silicon layer **212** may include a silicon wafer sliced from a silicon crystal ingot grown as a cylinder. The second silicon layer **212** may include an electrical conductivity value falling between that of a conductor, such as metallic copper, and an insulator, such as glass. The second silicon layer **212** may be replaced with other materials, such as germanium, gallium arsenide, silicon germanium, and/or the like.

(31) The doped layer **214** may include a material (e.g., silicon, germanium, silicon carbide, silicon germanium, and/or the like) that is doped with a dopant material (e.g., boron, arsenic, phosphorus, gallium, and/or the like). Doping is the intentional introduction of impurities into an intrinsic semiconductor material for the purpose of modulating electrical, optical, and/or structural properties of the semiconductor material. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. **1**, may be utilized to perform a deposition operation that forms the intrinsic semiconductor material on a top surface of the second silicon layer **212**. An ion implantation tool or a diffusion tool may also be utilized to implant the dopant material in the intrinsic semiconductor material to form the doped layer **214**.

(32) The epitaxial layer **216** may include a silicon germanium formed via epitaxial growth. In some implementations, the epitaxial layer **216** includes other materials, such as silicon, silicon carbide, silicon germanium, gallium arsenide, gallium phosphide, and/or the like. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. **1**, may be utilized to perform a deposition operation that forms the epitaxial layer **216** on a top surface of the doped layer **214**.

(33) The IMD layer **218** may include an intermetal dielectric material, such as silicon dioxide, a low dielectric constant (e.g., k values in a range of 3.2 to 2.0) dielectric material, fluorinated silica glass, silicon, silicon nitride, silicon oxide, and/or the like, that electrically insulates the second metal contacts **220** from other components of the second semiconductor device **210**. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. **1**, may be utilized to form the IMD layer **218** on a top surface of epitaxial layer **216**. For example, deposition tool **102** may perform a PECVD operation, an HDP-CVD operation, an SACVD operation, an ALD operation, a PVD operation, or another deposition operation to deposit the IMD layer **218** on the top surface of the epitaxial layer **216**.

(34) The second metal contacts **220** may include a conductive metal, such as titanium, cobalt, tungsten, aluminum, copper, ruthenium, iridium, and/or the like. In some implementations, the second metal contacts **220** may be formed within openings formed in the IMD layer **218**. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. **1**, may be utilized to perform a deposition operation that forms second metal contacts **220** in the openings of IMD layer **218**. In some implementations, a plating tool is used to form the second metal contacts **220** in the IMD layer **218**. In these examples, the plating tool may perform a plating operation to form the second metal contacts **220**. The plating operation may include applying a voltage across an anode formed of a plating material and a cathode (e.g., a substrate). The voltage causes a current to oxidize the anode, which causes the release of plating material ions from the anode. These plating material ions form a plating solution that travels through a plating bath toward the substrate (e.g., the IMD layer **218**). The plating solution reaches the IMD layer **218** and deposits plating material ions into the openings in the IMD layer **218** to form the second metal contacts **220**.

(35) In some implementations, one or more semiconductor processing tools of the environment

100, described above in connection with FIG. **1**, may be utilized to form the openings in the IMD layer **218**, prior to formation of the second metal contacts **220** in the openings. For example, the deposition tool **102** may form a photoresist layer on the the IMD layer **218**, the exposure tool **104** may expose the photoresist layer to a radiation source to pattern the photoresist layer, the developer tool **106** may develop and remove portions of the photoresist layer to expose the pattern, and the etching tool **108** may etch the one or more portions of the IMD layer **218** to form the openings in the USG layer **206**. In some implementations, a photoresist removal tool removes the remaining portions of the photoresist layer (e.g., using a chemical stripper and/or another technique) after the etching tool **108** etches the IMD layer **218**.

(36) The passivation layer **221** may include an oxide material (e.g., a metal oxide) that is inert and does not change semiconductor properties as a result of interaction with air or other materials in contact with the passivation layer **221**. The passivation layer **221** may allow electricity to reliably penetrate to conducting layers provided below the passivation layer **221**, and to overcome surface states that prevent electricity from reaching the conducting layers. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. **1**, may be utilized to form the passivation layer **221** on the top surface of the IMD layer **218** and the second metal contacts **220**.

(37) As shown in FIG. **2B**, and by reference number **222**, a deposition operation may be performed to form a first bonding layer **224** on a top surface of the first semiconductor device **202** and a second bonding layer **226** on a top surface of the second semiconductor device **210**. For example, the first bonding layer **224** may be formed on top surfaces of the USG layer **206** and the first metal contacts **208**, and the second bonding layer **226** may be formed on top surfaces of the IMD layer **218** and the second metal contacts **220**. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. **1**, may be utilized to form the first bonding layer **224** on the top surface of the first semiconductor device **202** and the second bonding layer **226** on the top surface of the second semiconductor device **210**. For example, the deposition tool **102** may perform a CVD operation, a PECVD operation, an HDP-CVD operation, an SACVD operation, an ALD operation, a PVD operation, or another deposition operation to form the first bonding layer **224** on the top surface of the first semiconductor device **202** and the second bonding layer **226** on the top surface of second semiconductor device **210**.

(38) In some implementations, a planarization operation may be performed on the first bonding layer **224** and/or the second bonding layer **226** to flatten the first bonding layer **224** and/or the second bonding layer **226**. In some implementations, the planarization tool **110** of the environment **100**, described above in connection with FIG. **1**, may perform the planarization operation. The planarization operation may include a chemical mechanical polishing/planarization (CMP) operation or another type of planarization operation. A CMP operation may include depositing a slurry (or polishing compound) onto a polishing pad. The first semiconductor device **202** may be mounted to a carrier, which may rotate the first semiconductor device **202** as the first semiconductor device **202** is pressed against the polishing pad. The slurry and polishing pad act as an abrasive that polishes or planarizes the first bonding layer **224** as the first semiconductor device **202** is rotated. The polishing pad may also be rotated to ensure a continuous supply of slurry is applied to the polishing pad. Similar techniques may be performed to planarize the second bonding layer **226** of the second semiconductor device **210**.

(39) In some implementations, the first bonding layer **224** and/or the second bonding layer **226** may be planarized to a particular thickness. For example, the first bonding layer **224** and/or the second bonding layer **226** may be planarized to a thickness in a range of approximately 10 angstroms to approximately 100,000 angstroms such that control over the surface uniformity and roughness may be maintained for the first bonding layer **224** and/or the second bonding layer **226**. In some implementations, the first bonding layer **224** and/or the second bonding layer **226** may be planarized to achieve a particular surface roughness. For example, the first bonding layer **224**

and/or the second bonding layer **226** may be planarized to achieve a surface roughness of less than 1 angstrom.

(40) The first bonding layer **224** and the second bonding layer **226** may be heterogeneous bonding layers. In particular, the first bonding layer **224** and the second bonding layer **226** may be formed of one or more materials such that a chemical composition of the first bonding layer **224** and a chemical composition of the second bonding layer **226** are different chemical compositions. The first bonding layer **224** may be formed of one or more materials such that the chemical composition of the first bonding layer **224** is high in silicon content and hydroxy-containing silicon content (e.g., hydroxy group content), and low in nitrogen content. The second bonding layer **226** may be formed of one or more materials such that the chemical composition of the second bonding layer **226** includes silicon and is high in nitrogen content.

(41) The silicon concentration (e.g., the hydroxy-containing silicon concentration) of the first bonding layer **224** may be greater than the silicon concentration of the second bonding layer **226**. The high silicon concentration (e.g., the hydroxy-containing silicon concentration) of the first bonding layer **224** may cause or facilitate the formation of silicon-oxygen-silicon bonds during an annealing operation to bond the first semiconductor device **202** and the second semiconductor device **210**. The nitrogen concentration of the second bonding layer **226** may be greater than the nitrogen concentration of the first bonding layer **224**. The high nitrogen concentration of the second bonding layer **226** may enhance and/or increase the ease of the formation of the silicon-oxygen-silicon bonds during the annealing operation.

(42) Accordingly, the silicon-to-nitrogen ratio of the first bonding layer **224** may be greater relative to the silicon-to-nitrogen ratio of the second bonding layer **226**. As an example, the silicon-to-nitrogen ratio of the first bonding layer **224** may be approximately 20 or greater, and the silicon-to-nitrogen ratio of the second bonding layer **226** may be approximately 2 or less to facilitate the formation of silicon-oxygen-silicon bonds during an annealing operation to bond the first semiconductor device **202** and the second semiconductor device **210**. As another example, the silicon-to-nitrogen ratio of the first bonding layer **224** may be in a range of approximately 20 to approximately 1000, and the silicon-to-nitrogen ratio of the second bonding layer **226** may be in a range of approximately 0.5 to approximately 2 to enhance and/or increase the formation of the silicon-oxygen-silicon bonds during the annealing operation.

(43) In some implementations, the first bonding layer **224** is formed of a silicon oxide (SiO) (e.g., a hydroxy-containing silicon oxide), a silicon dioxide (SiO₂) (e.g., a hydroxy-containing silicon dioxide), a silicon oxycarbide (SiOC) (e.g., a hydroxy-containing silicon oxycarbide), or another silicon-containing material. The silicon-containing material may be bonded to hydroxy (or hydroxy group) components such as OH to form Si—OH bonds in the first bonding layer **224**. In some implementations, the second bonding layer **226** is formed of a silicon nitride (SiN), a silicon oxynitride (SiON), a silicon carbon nitride (SiCN), or another silicon and nitrogen containing material.

(44) As shown in FIG. 2C, and by reference number **228**, an annealing operation may be performed to fuse the first bonding layer **224** and the second bonding layer **226**, which bonds the first semiconductor device **202** and the second semiconductor device **210**. For example, one of the first semiconductor device **202** or the second semiconductor device **210** may be rotated one-hundred and eighty degrees so that the first bonding layer **224** faces the second bonding layer **226**. FIG. 2C shows the second semiconductor device **210** being rotated one-hundred and eighty degrees, but the first semiconductor device **202** may be rotated one-hundred and eighty degrees instead of the second semiconductor device **210**. Once the first bonding layer **224** faces the second bonding layer **226**, the first bonding layer **224** may be bonded together with the second bonding layer **226**, which may join the first semiconductor device **202** and the second semiconductor device **210**. Thus, as shown in the example orientation of FIG. 2C, the second bonding layer **226** may be provided on a top surface of the first bonding layer **224**. The second metal contacts **220** and the IMD layer **218**

may be provided on the second bonding layer **226**, and the epitaxial layer **216** may be provided on the IMD layer **218**. The doped layer **214** may be provided on the epitaxial layer **216**, and the second silicon layer **212** may be provided on the doped layer **214**.

(45) A bonding strength of each of the first bonding layer **224** and the second bonding layer **226** may be greater than two Joules per square meter to enable direct bonding of the first semiconductor device **202** and the second semiconductor device **210**. For example, the bonding strength of each of the first bonding layer **224** and the second bonding layer **226** may be approximately greater than 2.5 Joules per square meter. The first semiconductor device **202** and the second semiconductor device **210** may be joined via the first bonding layer **224** and the second bonding layer **226** without pretreating the first bonding layer **224** and the second bonding layer **226** using a plasma treatment process. The bonding strength of the first bonding layer **224** and the second bonding layer **226** eliminates a need for expensive and time-consuming plasma pretreatment utilized in current bonding processes.

(46) In some implementations, the annealing tool **112** of the environment **100**, described above in connection with FIG. **1**, may be utilized to perform the annealing operation to fuse or bond the first bonding layer **224** and the second bonding layer **226** (e.g., through covalent bonding of the first bonding layer **224** and the second bonding layer **226**). In some implementations, the annealing operation may be performed under particular process conditions to fuse or bond the first bonding layer **224** and the second bonding layer **226**. For example, the annealing operation may be performed at a temperature in a range from approximately 150 degrees Celsius to approximately 400 degrees Celsius and for a time period in a range from approximately 30 minutes to approximately 3 hours, to permit the covalent bonds to form between the first bonding layer **224** and the second bonding layer **226**.

(47) In some implementations, the annealing operation is a dry anneal, an RTA, or another type of annealing operation. The annealing operation may result in a dehydration reaction that occurs between the first bonding layer **224** and the second bonding layer **226**. The dehydration reaction is a reaction that results in water (e.g., H_2O) being removed from the first bonding layer **224** and/or the second bonding layer **226**. The dehydration reaction causes the hydroxy components (e.g., the OH components) of the first bonding layer **224** to decompose, which forms silicon-oxygen-silicon bonds (e.g., Si—O—Si bonds) between the first bonding layer **224** and the second bonding layer **226**. Here, the silicon and the oxygen in silicon-hydroxy bonds (e.g., the Si—OH bonds) of the first bonding layer **224** are bonded with the silicon in the second bonding layer **226** to form the Si—O—Si bonds, with water forming as byproduct of the dehydration reaction. Moreover, the nitrogen in the second bonding layer **226** increases the effectiveness of the dehydration reaction and, therefore, the effectiveness and strength of the bond between the first bonding layer **224** and the second bonding layer **226**. In particular, the nitrogen content in the second bonding layer **226** permits the Si—O—Si bonds between the first bonding layer **224** and the second bonding layer **224** to be more easily formed (e.g., than in the absence of nitrogen) during the annealing operation.

(48) As shown in FIG. **2D**, and by reference number **230**, an etching operation may be performed to remove the second silicon layer **212** and the doped layer **214** from the epitaxial layer **216**. In some implementations, a first etching operation is performed to remove the second silicon layer **212** from the doped layer **214**, and a second etching operation is performed to remove the doped layer **214** from the epitaxial layer **216**. In some implementations, the etching tool **108** of the environment **100**, described above in connection with FIG. **1**, may be utilized to perform the first etching operation to remove the second silicon layer **212** from the doped layer **214**, and to perform the second etching operation to remove the doped layer **214** from the epitaxial layer **216**. In some implementations, a single etching operation is performed to remove the second silicon layer **212** and the doped layer **214** from the epitaxial layer **216**. In some implementations, the etching operation(s) may include wet etching operation(s), dry etching operation(s), plasma etching

operation(s), and/or other types of etching operation(s).

(49) As shown in FIG. 2E, and by reference number **232**, a passivation layer **234** may be formed on the epitaxial layer **216**. For example, the passivation layer **234** may be deposited on a top surface of the epitaxial layer **216**. The passivation layer **234** may include an oxide material (e.g., a metal oxide) that is inert and does not change semiconductor properties as a result of interaction with air or other materials in contact with the passivation layer **234**. The passivation layer **234** may allow electricity to reliably penetrate to conducting layers provided below the passivation layer **234**, and to overcome surface states that prevent electricity from reaching the conducting layers. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. 1, may be utilized to form the passivation layer **234** on the top surface of the epitaxial layer **216**.

(50) As shown in FIG. 2F, and by reference number **236**, metal vias **238** may be formed, through the passivation layer **234**, the epitaxial layer **216**, the IMD layer **218**, the first bonding layer **224**, and the second bonding layer **226**, to connect with the first metal contacts **208** and the second metal contacts **220**. For example, the deposition tool **102**, the exposure tool **104**, the developer tool **106**, and the etching tool **108** may be performed to form openings through the passivation layer **234**, the epitaxial layer **216**, the IMD layer **218**, the second metal contacts **220**, the first bonding layer **224**, and/or the second bonding layer **226**. The deposition tool **102** may perform one or more deposition operations, or a plating tool may perform one or more electroplating operations, to provide the metal vias **238** in the openings formed through the passivation layer **234**, the epitaxial layer **216**, the IMD layer **218**, the second metal contacts **220**, the first bonding layer **224**, and/or the second bonding layer **226**.

(51) The metal vias **238** may include a conductive metal, such as titanium, cobalt, tungsten, aluminum, copper, ruthenium, iridium, and/or the like. In some implementations, a metal via **238** may be formed for each of the first metal contacts **208** and each of the second metal contacts **220**. In some implementations, a single metal via **238** may be formed such that the single metal via **238** connects with two or more of the first metal contacts **208** and/or two or more of the second metal contacts **220**.

(52) As shown in FIG. 2G, and by reference number **240**, third metal contacts **242** may be formed on the metal vias **238**. The third metal contacts **242** may include a conductive metal, such as titanium, cobalt, tungsten, aluminum, copper, ruthenium, iridium, and/or the like. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. 1, may be utilized to form third metal contacts **246** on metal vias **238**. In some implementations, a plating tool may be used to perform an electroplating process to form the third metal contacts **246** on the metal vias **238**. The final arrangement of the device (or the portion thereof) may include stacked semiconductor devices (e.g., the first semiconductor device **202** and the second semiconductor device **210**), a three-dimensional integrated circuit, and/or the like. In some implementations, a third metal contact **242** may be formed on each of the metal vias **238**. In some implementations, a single third metal contact **242** may be formed on two or more of the metal vias **238**.

(53) The number and arrangement of structures, layers, and/or the like shown in FIGS. 2A-2G are provided as an example. In practice, the device may include additional structures and/or layers, fewer structures and/or layers, different structures and/or layers, or differently arranged structures and/or layers than those shown in FIG. 2A-2G.

(54) FIG. 3 is a diagram of example components of a device **300**. In some implementations, the deposition tool **102**, the exposure tool **104**, the developer tool **106**, the etching tool **108**, the planarization tool **110**, the annealing tool **112**, and/or wafer/die transport tool **114** may include one or more devices **300** and/or one or more components of device **300**. As shown in FIG. 3, device **300** may include a bus **310**, a processor **320**, a memory **330**, a storage component **340**, an input component **350**, an output component **360**, and a communication component **370**.

(55) Bus **310** includes a component that permits communication among the components of device **300**. Processor **320** is implemented in hardware, firmware, or a combination of hardware and software. Processor **320** is a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), a microprocessor, a microcontroller, a digital signal processor (DSP), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), or another type of processing component. In some implementations, processor **320** includes one or more processors capable of being programmed to perform a function. Memory **330** includes a random access memory (RAM), a read only memory (ROM), and/or another type of dynamic or static storage device (e.g., a flash memory, a magnetic memory, and/or an optical memory) that stores information and/or instructions for use by processor **320**.

(56) Storage component **340** stores information and/or software related to the operation and use of device **300**. For example, storage component **340** may include a hard disk (e.g., a magnetic disk, an optical disk, a magneto-optic disk, and/or a solid state disk), a compact disc (CD), a digital versatile disc (DVD), a floppy disk, a cartridge, a magnetic tape, and/or another type of non-transitory computer-readable medium, along with a corresponding drive.

(57) Input component **350** includes a component that permits device **300** to receive information, such as via user input (e.g., a touch screen display, a keyboard, a keypad, a mouse, a button, a switch, and/or a microphone). Additionally, or alternatively, input component **350** may include a sensor for sensing information (e.g., a global positioning system (GPS) component, an accelerometer, a gyroscope, and/or an actuator). Output component **360** includes a component that provides output information from device **300** (e.g., a display, a speaker, and/or one or more LEDs).

(58) Communication component **370** includes a transceiver-like component (e.g., a transceiver and/or a separate receiver and transmitter) that enables device **300** to communicate with other devices, such as via a wired connection, a wireless connection, or a combination of wired and wireless connections. Communication component **370** may permit device **300** to receive information from another device and/or provide information to another device. For example, communication component **370** may include an Ethernet interface, an optical interface, a coaxial interface, an infrared interface, a radio frequency (RF) interface, a universal serial bus (USB) interface, a wireless local area interface, a cellular network interface, and/or the like.

(59) Device **300** may perform one or more processes described herein. Device **300** may perform these processes based on processor **320** executing software instructions stored by a non-transitory computer-readable medium, such as memory **330** and/or storage component **340**. A computer-readable medium is defined herein as a non-transitory memory device. A memory device includes memory space within a single physical storage device or memory space spread across multiple physical storage devices.

(60) Software instructions may be read into memory **330** and/or storage component **340** from another computer-readable medium or from another device via communication component **370**. When executed, software instructions stored in memory **330** and/or storage component **340** may cause processor **320** to perform one or more processes described herein. Additionally, or alternatively, hardwired circuitry may be used in place of or in combination with software instructions to perform one or more processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(61) The number and arrangement of components shown in FIG. 3 are provided as an example. In practice, device **300** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. 3. Additionally, or alternatively, a set of components (e.g., one or more components) of device **300** may perform one or more functions described as being performed by another set of components of device **300**.

(62) FIG. 4 is a flowchart of an example process **400** associated with joining semiconductor devices. In some implementations, one or more process blocks of FIG. 4 may be performed by one or more semiconductor processing tool (e.g., one or more of the semiconductor processing tools

102-110). Additionally, or alternatively, one or more process blocks of FIG. 4 may be performed by one or more components of device **300**, such as processor **320**, memory **330**, storage component **340**, input component **350**, output component **360**, and/or communication component **370**.

(63) As shown in FIG. 4, process **400** may include forming a first bonding layer on a surface of a first semiconductor device (block **410**). For example, a semiconductor processing tool (e.g., the deposition tool **102**) may form a first bonding layer **224** on a surface of a first semiconductor device **202**, as described above.

(64) As further shown in FIG. 4, process **400** may include forming a second bonding layer on a surface of a second semiconductor device, wherein a chemical composition of the first bonding layer is different from a chemical composition of the second bonding layer to promote bonding between the first semiconductor device and the second semiconductor device (block **420**). For example, a semiconductor processing tool (e.g., the deposition tool **102**) may form a second bonding layer **226** on a surface of a second semiconductor device **210**, as described above. In some implementations, a chemical composition of the first bonding layer **224** is different from a chemical composition of the second bonding layer **226** to promote bonding between the first semiconductor device **202** and the second semiconductor device **210**.

(65) As further shown in FIG. 4, process **400** may include performing an annealing operation to bond the first semiconductor device with the second semiconductor device via the first bonding layer and the second bonding layer (block **430**). For example, the semiconductor processing tool (e.g., the annealing tool **112**) may perform an annealing operation to bond the first semiconductor device **202** with the second semiconductor device **210** via the first bonding layer **224** and the second bonding layer **226**, as described above.

(66) Process **400** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

(67) In a first implementation, the first bonding layer **224** and the second bonding layer **226** each includes a thickness in a range from approximately 10 angstroms to approximately 100,000 angstroms. In a second implementation, alone or in combination with the first implementation, a silicon-to-nitrogen ratio of the chemical composition of the first bonding layer **224** is greater relative to a silicon-to-nitrogen ratio of the chemical composition of the second bonding layer **226**. In a third implementation, alone or in combination with one or more of the first and second implementations, the chemical composition of the first bonding layer comprises a hydroxy-containing silicon dioxide, and wherein the chemical composition of the second bonding layer comprises a silicon nitride. In a fourth implementation, alone or in combination with one or more of the first through third implementations, performing the annealing operation to bond the first semiconductor device **202** with the second semiconductor device **210** via the first bonding layer **224** and the second bonding layer **226** includes performing the annealing operation to bond the first semiconductor device **202** with the second semiconductor device **210** via the first bonding layer **224** and the second bonding layer **226** without pretreating the first bonding layer or the second bonding layer.

(68) In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, performing the annealing operation fuse the first bonding layer **224** and the second bonding layer **226**, where the chemical composition of the first bonding layer **224** and the chemical composition of the second bonding layer **226** cause formation of silicon-oxygen-silicon bonds between the first bonding layer **224** and the second bonding layer **226** to directly bond the first semiconductor device **202** and the second semiconductor device **210**. In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, a roughness of the first bonding layer **224** or the second bonding layer **226** is less than approximately 1 angstrom.

(69) Although FIG. 4 shows example blocks of process **400**, in some implementations, process **400**

may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 4. Additionally, or alternatively, two or more of the blocks of process **400** may be performed in parallel.

(70) FIG. 5 is a flowchart of an example process **500** associated with bonding semiconductor devices. In some implementations, one or more process blocks of FIG. 5 may be performed by one or more semiconductor processing tool (e.g., one or more of the semiconductor processing tools **102-110**). Additionally, or alternatively, one or more process blocks of FIG. 5 may be performed by one or more components of device **300**, such as processor **320**, memory **330**, storage component **340**, input component **350**, output component **360**, and/or communication component **370**.

(71) As shown in FIG. 5, process **500** may include forming a first bonding layer on a surface of a first semiconductor device (block **510**). For example, a semiconductor processing tool (e.g., the deposition tool **102**) may form a first bonding layer **224** on a surface of a first semiconductor device **202**, as described above.

(72) As further shown in FIG. 5, process **500** may include forming a second bonding layer on a surface of a second semiconductor device (block **520**). For example, a semiconductor processing tool (e.g., the deposition tool **102**) may form a second bonding layer **226** on a surface of a second semiconductor device **210**, as described above.

(73) As further shown in FIG. 5, process **500** may include performing an annealing operation to bond the first semiconductor device and the second semiconductor device via the first bonding layer and the second bonding layer, wherein a hydroxy-containing silicon concentration of the first bonding layer results in formation of silicon-oxygen-silicon bonds during the annealing operation, and wherein a nitrogen concentration of the second bonding layer increases the formation of the silicon-oxygen-silicon bonds during the annealing operation (block **530**). For example, a semiconductor processing tool (e.g., the annealing tool **112**) may perform an annealing operation to bond the first semiconductor device **202** and the second semiconductor device **210** via the first bonding layer **224** and the second bonding layer **226**, as described above. In some implementations, a hydroxy-containing silicon concentration of the first bonding layer **224** results in formation of silicon-oxygen-silicon bonds during the annealing operation. In some implementations, a nitrogen concentration of the second bonding layer **226** increases the formation of the silicon-oxygen-silicon bonds during the annealing operation.

(74) Process **500** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

(75) In a first implementation, a silicon-to-nitrogen ratio of the first bonding layer **224** is approximately 20 or greater. In a second implementation, alone or in combination with the first implementation, a silicon-to-nitrogen ratio of the second bonding layer **226** is approximately 2 or less. In a third implementation, alone or in combination with one or more of the first and second implementations, the nitrogen concentration of the second bonding layer increases effectiveness of a dehydration reaction that occurs between the first bonding layer and the second bonding layer during the annealing operation.

(76) In a fourth implementation, alone or in combination with one or more of the first through third implementations, performing the annealing operation to bond the first semiconductor device **202** and the second semiconductor device **210** via the first bonding layer **224** and the second bonding layer **226** includes performing the annealing operation to bond the first semiconductor device **202** and the second semiconductor device **210** via the first bonding layer **224** and the second bonding layer **226** without performing a pretreatment process for the first bonding layer **224** or the second bonding layer **226** prior to the annealing operation. In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, performing the annealing operation to bond the first semiconductor device **202** and the second semiconductor device **210** via the first bonding layer **224** and the second bonding layer **226** includes performing

the annealing operation, at a temperature in a range from approximately 150 degrees Celsius to approximately 400 degrees Celsius and for a time period in a range from approximately 30 minutes to approximately 3 hours, to fuse the first bonding layer **224** and the second bonding layer **226**. (77) Although FIG. 5 shows example blocks of process **500**, in some implementations, process **500** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 5. Additionally, or alternatively, two or more of the blocks of process **500** may be performed in parallel.

(78) FIGS. 6A-6D are diagrams of one or more example operations **600** involved in manufacturing an example semiconductor device described herein. In particular, the one or more example operations **600** may be performed to bond the second semiconductor device **210** to a carrier substrate such as a silicon carrier wafer such that one or more additional operations (e.g., one or more backside operations) may be performed on the second semiconductor device **210** while the second semiconductor device **210** is supported by the carrier substrate.

(79) As shown in FIG. 6A, a first semiconductor device **202** may include a first silicon layer **204**. The first silicon layer **204** may be a carrier substrate such as a silicon carrier wafer. As further shown in FIG. 6A, second semiconductor device **210** may include the second silicon layer **212**, the doped layer **214** provided on the second silicon layer **212**, the epitaxial layer **216** formed on the doped layer **214**, the IMD layer **218** formed on the epitaxial layer **216**, and the second metal contacts **220** formed in the IMD layer **218**. In some implementations, the passivation layer **221** may also be formed on the IMD layer **218** and the second metal contacts **220**.

(80) As shown in FIG. 6B, and by reference number **602**, a deposition operation may be performed to form a first bonding layer **224** on a top surface of the first semiconductor device **202** and a second bonding layer **226** on a top surface of the second semiconductor device **210**. For example, the first bonding layer **224** may be formed on the top surface of the first silicon layer **204**, and the second bonding layer **226** may be formed on top surfaces of the IMD layer **218** and the second metal contacts **220**. In some implementations, the deposition tool **102** of the environment **100**, described above in connection with FIG. 1, may be utilized to form the first bonding layer **224** on the top surface of the first semiconductor device **202** and the second bonding layer **226** on the top surface of the second semiconductor device **210**. For example, the deposition tool **102** may perform a CVD operation, a PECVD operation, an HDP-CVD operation, an SACVD operation, an ALD operation, a PVD operation, or another deposition operation to form the first bonding layer **224** on the top surface of the first semiconductor device **202** and the second bonding layer **226** on the top surface of second semiconductor device **210**.

(81) In some implementations, a planarization operation may be performed on the first bonding layer **224** and/or the second bonding layer **226** to flatten the first bonding layer **224** and/or the second bonding layer **226**. In some implementations, the planarization tool **110** of the environment **100**, described above in connection with FIG. 1, may perform the planarization operation. The planarization operation may include a CMP operation or another type of planarization operation. In some implementations, the first bonding layer **224** and/or the second bonding layer **226** may be planarized to a particular thickness. For example, the first bonding layer **224** and/or the second bonding layer **226** may be planarized to a thickness in a range of approximately 10 angstroms to approximately 100,000 angstroms such that control over the surface uniformity and roughness may be maintained for the first bonding layer **224** and/or the second bonding layer **226**. In some implementations, the first bonding layer **224** and/or the second bonding layer **226** may be planarized to achieve a particular surface roughness. For example, the first bonding layer **224** and/or the second bonding layer **226** may be planarized to achieve a surface roughness of less than 1 angstrom.

(82) The first bonding layer **224** and the second bonding layer **226** may be heterogeneous bonding layers. In particular, the first bonding layer **224** and the second bonding layer **226** may be formed of one or more materials such that a chemical composition of the first bonding layer **224** and a

chemical composition of the second bonding layer **226** are different chemical compositions. The first bonding layer **224** may be formed of one or more materials such that the chemical composition of the first bonding layer **224** is high in silicon content and hydroxy-containing silicon content (e.g., hydroxy group content), and low in nitrogen content. The second bonding layer **226** may be formed of one or more materials such that the chemical composition of the second bonding layer **226** includes silicon and is high in nitrogen content.

(83) The silicon concentration (e.g., the hydroxy-containing silicon concentration) of the first bonding layer **224** may be greater than the silicon concentration of the second bonding layer **226**. The high silicon concentration (e.g., the hydroxy-containing silicon concentration) of the first bonding layer **224** may cause or facilitate the formation of silicon-oxygen-silicon bonds during an annealing operation to bond the first semiconductor device **202** and the second semiconductor device **210**. The nitrogen concentration of the second bonding layer **226** may be greater than the nitrogen concentration of the first bonding layer **224**. The high nitrogen concentration of the second bonding layer **226** may enhance and/or increase the ease of the formation of the silicon-oxygen-silicon bonds during the annealing operation.

(84) Accordingly, the silicon-to-nitrogen ratio of the first bonding layer **224** may be greater relative to the silicon-to-nitrogen ratio of the second bonding layer **226**. As an example, the silicon-to-nitrogen ratio of the first bonding layer **224** may be approximately 20 or greater, and the silicon-to-nitrogen ratio of the second bonding layer **226** may be approximately 2 or less to facilitate the formation of silicon-oxygen-silicon bonds during an annealing operation to bond the first semiconductor device **202** and the second semiconductor device **210**. As another example, the silicon-to-nitrogen ratio of the first bonding layer **224** may be in a range of approximately 20 to approximately 1000, and the silicon-to-nitrogen ratio of the second bonding layer **226** may be in a range of approximately 0.5 to approximately 2 to enhance and/or increase the formation of the silicon-oxygen-silicon bonds during the annealing operation.

(85) In some implementations, the first bonding layer **224** is formed of a silicon oxide (SiO) (e.g., a hydroxy-containing silicon oxide), a silicon dioxide (SiO₂) (e.g., a hydroxy-containing silicon dioxide), a silicon oxycarbide (SiOC) (e.g., a hydroxy-containing silicon oxycarbide), or another silicon-containing material. The silicon-containing material may be bonded to hydroxy (or hydroxy group) components such as OH to form Si—OH bonds in the first bonding layer **224**. In some implementations, the second bonding layer **226** is formed of a silicon nitride (SiN), a silicon oxynitride (SiON), a silicon carbon nitride (SiCN), or another silicon and nitrogen containing material.

(86) As shown in FIG. **6C**, and by reference number **604**, an annealing operation may be performed to fuse the first bonding layer **224** and the second bonding layer **226**, which bonds the first semiconductor device **202** and the second semiconductor device **210**. For example, one of the first semiconductor device **202** or the second semiconductor device **210** may be rotated one-hundred and eighty degrees so that the first bonding layer **224** faces the second bonding layer **226**. FIG. **6C** shows the second semiconductor device **210** being rotated one-hundred and eighty degrees, but the first semiconductor device **202** may be rotated one-hundred and eighty degrees instead of the second semiconductor device **210**. Once the first bonding layer **224** faces the second bonding layer **226**, the first bonding layer **224** may be bonded together with the second bonding layer **226**, which may join the first semiconductor device **202** and the second semiconductor device **210**. Thus, as shown in the example orientation of FIG. **6C**, the second bonding layer **226** may be provided on a top surface of the first bonding layer **224**. The second metal contacts **220** and the IMD layer **218** may be provided on the second bonding layer **226**, and the epitaxial layer **216** may be provided on the IMD layer **218**. The doped layer **214** may be provided on the epitaxial layer **216**, and the second silicon layer **212** may be provided on the doped layer **214**.

(87) A bonding strength of each of the first bonding layer **224** and the second bonding layer **226** may be greater than two Joules per square meter to enable direct bonding of the first semiconductor

device **202** and the second semiconductor device **210**. For example, the bonding strength of each of the first bonding layer **224** and the second bonding layer **226** may be approximately greater than 2.5 Joules per square meter. The first semiconductor device **202** and the second semiconductor device **210** may be joined via the first bonding layer **224** and the second bonding layer **226** without pretreating the first bonding layer **224** and the second bonding layer **226** using a plasma treatment process. The bonding strength of the first bonding layer **224** and the second bonding layer **226** eliminates a need for expensive and time-consuming plasma pretreatment utilized in current bonding processes.

(88) In some implementations, the annealing tool **112** of the environment **100**, described above in connection with FIG. **1**, may be utilized to perform the annealing operation to fuse or bond the first bonding layer **224** and the second bonding layer **226** (e.g., through covalent bonding of the first bonding layer **224** and the second bonding layer **226**). In some implementations, the annealing operation may be performed under particular process conditions to fuse or bond the first bonding layer **224** and the second bonding layer **226**. For example, the annealing operation may be performed at a temperature in a range from approximately 150 degrees Celsius to approximately 400 degrees Celsius and for a time period in a range from approximately 30 minutes to approximately 3 hours, to permit the covalent bonds to form between the first bonding layer **224** and the second bonding layer **226**.

(89) In some implementations, the annealing operation is a dry anneal, an RTA, or another type of annealing operation. The annealing operation may result in a dehydration reaction that occurs between the first bonding layer **224** and the second bonding layer **226**. The dehydration reaction is a reaction that results in water (e.g., H.sub.2O) being removed from the first bonding layer **224** and/or the second bonding layer **226**. The dehydration reaction causes the hydroxy components (e.g., the OH components) of the first bonding layer **224** to decompose, which forms silicon-oxygen-silicon bonds (e.g., Si—O—Si bonds) between the first bonding layer **224** and the second bonding layer **226**. Here, the silicon and the oxygen in silicon-hydroxy bonds (e.g., the Si—OH bonds) of the first bonding layer **224** are bonded with the silicon in the second bonding layer **226** to form the Si—O—Si bonds, with water forming as byproduct of the dehydration reaction. Moreover, the nitrogen in the second bonding layer **226** increases the effectiveness of the dehydration reaction and, therefore, the effectiveness and strength of the bond between the first bonding layer **224** and the second bonding layer **226**. In particular, the nitrogen content in the second bonding layer **226** permits the Si—O—Si bonds between the first bonding layer **224** and the second bonding layer **224** to be more easily formed (e.g., than in the absence of nitrogen) during the annealing operation.

(90) As shown in FIG. **6D**, and by reference number **606**, an etching operation may be performed to remove the second silicon layer **212** and the doped layer **214** from the epitaxial layer **216**. In some implementations, a first etching operation is performed to remove the second silicon layer **212** from the doped layer **214**, and a second etching operation is performed to remove the doped layer **214** from the epitaxial layer **216**. In some implementations, the etching tool **108** of the environment **100**, described above in connection with FIG. **1**, may be utilized to perform the first etching operation to remove the second silicon layer **212** from the doped layer **214**, and to perform the second etching operation to remove the doped layer **214** from the epitaxial layer **216**. In some implementations, a single etching operation is performed to remove the second silicon layer **212** and the doped layer **214** from the epitaxial layer **216**. In some implementations, the etching operation(s) may include wet etching operation(s), dry etching operation(s), plasma etching operation(s), and/or other types of etching operation(s).

(91) In this way, the annealing operation described above in connection with reference number **606** may be performed to bond second semiconductor device **210** to a carrier substrate such as a silicon carrier wafer (e.g., first silicon layer **204**) such that one or more additional operations (e.g., one or more backside operations) may be performed on second semiconductor device **210** while second

semiconductor device **210** is supported by the carrier substrate.

(92) As indicated above, FIGS. **6A-6D** are provided merely as one or more examples. Other examples may differ from what is described with regard to FIGS. **6A-6D**.

(93) FIG. **7** is a flow chart of an example process **700** for joining semiconductor devices. In some implementations, one or more process blocks of FIG. **7** may be performed by a device (e.g., one or more of the tools depicted in FIG. **1**). In some implementations, one or more process blocks of FIG. **7** may be performed by another device or a group of devices separate from or including the one or more tools depicted in FIG. **1**. Additionally, or alternatively, one or more process blocks of FIG. **7** may be performed by one or more components of device **300**, such as processor **320**, memory **330**, storage component **340**, input component **350**, output component **360**, communication component **370**, and/or the like.

(94) As shown in FIG. **7**, process **700** may include forming a first bonding layer over a carrier substrate of a first semiconductor device (block **710**). For example, the device may form first bonding layer **224** over a carrier substrate of a first semiconductor device **202**, as described above. The carrier substrate includes a first silicon layer **204** such as a silicon carrier wafer.

(95) As further shown in FIG. **7**, process **700** may include forming a second bonding layer on a surface of a second semiconductor device (block **720**). For example, the device may form a second bonding layer **226** on a surface of a second semiconductor device **210**, as described above.

(96) As further shown in FIG. **7**, process **700** may include performing an annealing operation to bond the first semiconductor device with the second semiconductor device via the first bonding layer and the second bonding layer (block **730**). For example, the device may perform an annealing operation to bond the first semiconductor device **202** with the second semiconductor device **210** via the first bonding layer **224** and the second bonding layer **224**, as described above. In some implementations, a hydroxy-containing silicon concentration of the first bonding layer **224** results in formation of silicon-oxygen-silicon bonds during the annealing operation. In some implementations, a nitrogen concentration of the second bonding layer **226** increases the formation of the silicon-oxygen-silicon bonds during the annealing operation.

(97) As further shown in FIG. **7**, process **700** may include removing a silicon layer and a doped layer from the second semiconductor device after performing the annealing operation (block **740**). For example, the device may remove the second silicon layer **212** and a doped layer **214** from the second semiconductor device **210** after performing the annealing operation, as described above.

(98) Process **700** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein. Although FIG. **7** shows example blocks of process **700**, in some implementations, process **700** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **6**. Additionally, or alternatively, two or more of the blocks of process **700** may be performed in parallel.

(99) In this way, a first semiconductor device and a second semiconductor device may be directly bonded using heterogeneous bonding layers. A first bonding layer may be formed on the first semiconductor device and the second bonding layer may be formed on the second semiconductor device that is to be bonded or joined to the first semiconductor device. Each bonding layer may include a silicon-containing material. The first bonding layer may include a higher concentration of hydroxy-containing silicon and a lower concentration of nitrogen relative to the second bonding layer. The second bonding layer may include silicon with a higher concentration of nitrogen relative to the first bonding layer. A dry anneal may be performed to cause a dehydration reaction that results in decomposition of the hydroxy components of the first bonding layer, which forms silicon oxide bonds between the first bonding layer and the second bonding layer. The nitrogen in the second bonding layer increases the effectiveness of the dehydration reaction and, therefore, the effectiveness and strength of the bond between the first bonding layer and the second bonding layer. Accordingly, the first semiconductor device and the second semiconductor device may be

directly bonded without the use of a plasma pretreatment process, which decreases a cost and a complexity of the bonding process.

(100) As described in greater detail above, some implementations described herein provide a method. The method includes forming a first bonding layer on a surface of a first semiconductor device. The method includes forming a second bonding layer on a surface of a second semiconductor device. A chemical composition of the first bonding layer is different from a chemical composition of the second bonding layer to promote bonding between the first semiconductor device and the second semiconductor device. The method includes performing an annealing operation to bond the first semiconductor device with the second semiconductor device via the first bonding layer and the second bonding layer.

(101) As described in greater detail above, some implementations described herein provide a device. The device includes a first semiconductor device, a second semiconductor device, and a first bonding layer and a second bonding layer. The first bonding layer and the second bonding layer both join the first semiconductor device and the second semiconductor device. A chemical composition of the first bonding layer includes a greater silicon concentration relative to a chemical composition of the second bonding layer to cause formation of silicon-oxygen-silicon bonds between the first bonding layer and the second bonding layer. The chemical composition of the second bonding layer includes a greater nitrogen concentration relative to the chemical composition of the first bonding layer to increase the formation of the silicon-oxygen-silicon bonds between the first bonding layer and the second bonding layer.

(102) As described in greater detail above, some implementations described herein provide a method. The method includes forming a first bonding layer on a surface of a first semiconductor device. The method includes forming a second bonding layer on a surface of a second semiconductor device. The method includes performing an annealing operation to bond the first semiconductor device and the second semiconductor device via the first bonding layer and the second bonding layer. A hydroxy-containing silicon concentration of the first bonding layer results in formation of silicon-oxygen-silicon bonds during the annealing operation. A nitrogen concentration of the second bonding layer increases the formation of the silicon-oxygen-silicon bonds during the annealing operation.

(103) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device, comprising: a first semiconductor device; a second semiconductor device; and a first bonding layer and a second bonding layer, both joining the first semiconductor device and the second semiconductor device, wherein a chemical composition of the first bonding layer includes a greater silicon concentration relative to a chemical composition of the second bonding layer to cause formation of silicon-oxygen-silicon bonds between the first bonding layer and the second bonding layer, and wherein the chemical composition of the second bonding layer includes a greater nitrogen concentration relative to the chemical composition of the first bonding layer to increase the formation of the silicon-oxygen-silicon bonds between the first bonding layer and the second bonding layer.

2. The device of claim 1, wherein a silicon-to-nitrogen ratio of the chemical composition of the first

bonding layer is in a range of approximately 20 to approximately 1000.

3. The device of claim 1, wherein a silicon-to-nitrogen ratio of the chemical composition of the second bonding layer is in a range of approximately 0.5 to approximately 2.

4. The device of claim 1, wherein the first bonding layer includes at least one of: silicon oxide, silicon dioxide, or a silicon oxycarbide.

5. The device of claim 1, wherein the second bonding layer includes at least one of: a silicon nitride, a silicon carbon nitride, or a silicon oxynitride.

6. The device of claim 1, wherein a bond between the first bonding layer and the second bonding layer comprises a silicon-oxygen-silicon bond.

7. The device of claim 1, wherein a bonding strength between the first bonding layer and the second bonding layer is greater than two Joules per square meter to enable direct bonding of the first semiconductor device and the second semiconductor device.

8. A device, comprising: a first semiconductor device; a first bonding layer on a top surface of the first semiconductor device; a second semiconductor device; and a second bonding layer on a top surface of the second semiconductor device, wherein the first bonding layer is bonded to the second bonding layer, and wherein at least one of the first bonding layer or the second bonding layer has a surface roughness of less than 1 angstrom.

9. The device of claim 8, wherein at least one of: the first bonding layer is on a top surface of a first passivation layer of the first semiconductor device, or the second bonding layer is on a top surface of a second passivation layer of the second semiconductor device.

10. The device of claim 8, wherein at least one of the first bonding layer or the second bonding layer has a bonding strength of 2.5 Joules per square meter.

11. The device of claim 8, wherein at least one of the first bonding layer or the second bonding layer does not contain water.

12. The device of claim 8, wherein each of the first semiconductor device and the second semiconductor device comprises metal contacts.

13. The device of claim 12, wherein at least one metal contact, of metal contacts of the first semiconductor device, resides over at least one metal contact of the metal contacts of the second semiconductor device.

14. The device of claim 12, wherein at least one of: the metal contacts of the first semiconductor device reside in an undoped silicate glass (USG) layer of the first semiconductor device, or the metal contacts of the second semiconductor device reside in an intermetal dielectric (IMD) layer of the second semiconductor device.

15. A device, comprising: a first semiconductor device comprising a first passivation layer; a first bonding layer on a top surface of the first passivation layer; a second semiconductor device comprising a second passivation layer; a second bonding layer, on a top surface of the second passivation layer, bonded to the first bonding layer; and metal vias formed through the first passivation layer, the first bonding layer, the second passivation layer, and the second bonding layer, wherein at least one of the followings: the metal vias are formed through a portion of n undoped silicate glass (USG) layer of the first semiconductor device, the metal vias are formed through s portion of an intermetal dielectric (IMD) layer of the second semiconductor device, or the metal vias are further formed through a third passivation layer on the second semiconductor device.

16. The device of claim 15, wherein the first semiconductor device comprises first metal contacts and the second semiconductor device comprises second metal contacts, and wherein the metal vias are further formed through the first metal contacts and the second metal contacts.

17. The device of claim 15, wherein the metal vias are formed through the portion of the USG layer.

18. The device of claim 15, wherein the metal vias are formed through the portion of the IMD layer.

19. The device of claim 15, wherein the metal vias are further formed through the third passivation

layer.

20. The device of claim 19, further comprising: third metal contacts on the third passivation layer.
